



PHARMACOLOGY SUBJECT WISE TEST DATED ON 14th MAY 2026

QUESTIONS

- Q1. A 56-year-old man with a 10-year history of type 2 diabetes mellitus presents for follow-up. His blood pressure remains elevated at 150/94 mmHg despite adherence to lifestyle measures. Laboratory evaluation reveals persistent proteinuria and mildly raised serum creatinine. Fundoscopy shows early microvascular changes. The treating physician plans to initiate a drug that specifically reduces intraglomerular pressure and delays progression of diabetic nephropathy. Which of the following agents is most appropriate in this setting?
- a) Amlodipine
 - b) Metoprolol
 - c) Spironolactone
 - d) Telmisartan
- Q2. A 28-year-old woman presents with weight loss, palpitations, irritability, and heat intolerance for 2 months. On examination, she has tachycardia, fine tremors, and warm moist skin. Thyroid function tests confirm hyperthyroidism. She is started on a drug that inhibits thyroid hormone synthesis by blocking iodination and coupling reactions mediated by thyroid peroxidase. Which of the following drugs has this mechanism of action?
- a) Amiodarone
 - b) Propranolol
 - c) Methimazole
 - d) Lorazepam
- Q3. A 65-year-old male with a history of stable angina is prescribed aspirin. Shortly after ingestion, he develops wheezing, dyspnea, and bronchospasm suggestive of aspirin hypersensitivity. He also has a past history of nasal polyps. The physician decides to replace aspirin with an alternative antiplatelet agent that acts by inhibiting ADP-mediated platelet aggregation. Which of the following is the most suitable substitute?
- a) Heparin
 - b) Warfarin
 - c) Clopidogrel
 - d) Streptokinase
- Q4. A 52-year-old man with type 2 diabetes mellitus is started on an oral hypoglycemic drug that enhances insulin secretion by closing ATP-sensitive potassium channels in pancreatic β -cells. Within a few days, he reports episodes of sweating, dizziness, and confusion, particularly when meals are delayed. His symptoms improve after consuming glucose. Which of the following adverse effects is most likely associated with this class of drugs?
- a) Weight loss
 - b) Severe hypoglycemia



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- c) Lactic acidosis
- d) Genital infections

Q5. A 47-year-old male presents with poorly controlled hypertension despite multiple medications. He complains of muscle weakness and fatigue. Laboratory investigations reveal hypokalemia, metabolic alkalosis, and elevated plasma aldosterone levels. The physician adds a drug that selectively blocks aldosterone receptors without significant antiandrogenic effects. Which of the following is the most appropriate drug in this case?

- a) Amlodipine
- b) Nitroglycerin
- c) Eplerenone
- d) Furosemide

Q6. A hospitalized patient is diagnosed with a severe MRSA infection following a surgical procedure. Blood cultures confirm gram-positive cocci resistant to methicillin. He is started on a cephalosporin that is uniquely effective against MRSA due to its ability to bind altered penicillin-binding proteins. Which of the following drugs is most likely being used?

- a) Cefazolin
- b) Aztreonam
- c) Ceftriaxone
- d) Ceftobiprole

Q7. A patient with severe COVID-19 infection develops persistent high-grade fever, elevated CRP, ferritin, and IL levels, along with worsening respiratory distress. A diagnosis of cytokine storm is considered. He is treated with a monoclonal antibody that inhibits a key interleukin responsible for systemic inflammation and acute phase response. Which cytokine is targeted by this therapy?

- a) TNF- α
- b) IL-1
- c) CD-20
- d) IL-6

Q8. A 45-year-old woman with rheumatoid arthritis presents with progressive blurring of vision. Ophthalmological examination reveals corneal deposits and retinal pigmentary changes. She has been on long-term disease-modifying therapy. Which of the following drugs is most likely responsible for these ocular findings?

- a) Methotrexate
- b) Sulfasalazine
- c) Leflunomide
- d) Chloroquine



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Q9. A 60-year-old man with atrial fibrillation requires antiarrhythmic therapy. He has a known history of interstitial lung disease. The physician avoids prescribing a drug associated with pulmonary fibrosis as a serious adverse effect. Which of the following drugs is contraindicated in this patient?

- a) Lidocaine
- b) Quinidine
- c) Amiodarone
- d) Sotalol

Q10. A 62-year-old patient with chronic kidney disease presents with fatigue, pallor, and reduced exercise tolerance. Laboratory findings show normocytic normochromic anemia with low erythropoietin levels. Iron studies are within normal limits. The physician plans treatment targeting the underlying cause of anemia. Which of the following is the most appropriate therapy?

- a) Ferrous sulfate
- b) Sargramostim
- c) Romiplostim
- d) Darbepoetin alfa

Q11. A 64-year-old man with exercise-induced asthma is started on treatment for glaucoma. Shortly after initiating therapy, he develops worsening breathlessness and wheezing. The physician suspects that the prescribed drug has precipitated bronchospasm. Which of the following drugs is most likely responsible and should be avoided in such patients?

- a) Mannitol
- b) Acetazolamide
- c) Timolol
- d) Apraclonidine

Q12. A patient presents with fever, dry cough, and atypical pneumonia caused by *Mycoplasma pneumoniae*. Since the organism lacks a cell wall, certain antibiotics are ineffective. Which of the following drugs would NOT be useful in treating this infection?

- a) Azithromycin
- b) Doxycycline
- c) Moxifloxacin
- d) Vancomycin



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- Q13. A diabetic patient is started on combination therapy to improve glycemic control. After a few days, he develops episodes of sweating, confusion, and tremors, especially during fasting periods. Blood glucose during these episodes is significantly low. Which of the following drug combinations is most likely responsible for these hypoglycemic episodes?
- a) Insulin glargine + Metformin
 - b) Insulin degludec + Acarbose
 - c) Insulin lispro + Glimepiride
 - d) Insulin detemir + Pioglitazone
- Q14. A 30-year-old male presents with polyuria, polydipsia, and dehydration following a head injury. Laboratory evaluation reveals hypernatremia and low urine osmolality. Central diabetes insipidus is suspected. Which of the following is the most appropriate treatment in this condition?
- a) Tolvaptan
 - b) Hydrochlorothiazide
 - c) Insulin
 - d) Desmopressin
- Q15. A postoperative patient develops flushing, sweating, bradycardia, hypotension, and miosis after receiving a certain drug. These features are consistent with excessive parasympathetic stimulation. Which of the following best describes the mechanism of action of this drug?
- a) Muscarinic antagonist
 - b) Acetylcholinesterase inhibitor
 - c) μ -opioid receptor agonist
 - d) Muscarinic agonist
- Q16. A 68-year-old male presents with urinary hesitancy, weak stream, and nocturia. He is diagnosed with benign prostatic hyperplasia. A drug is prescribed that selectively blocks α_1A receptors in the prostate, improving urinary flow with minimal effect on systemic blood pressure. Which of the following is most appropriate?
- a) Terazosin
 - b) Phentolamine
 - c) Doxazosin
 - d) Tamsulosin
- Q17. A patient on long-term pharmacotherapy develops gynecomastia. The physician suspects a drug-induced cause and reviews the medication list. Which of the following drugs is least likely to cause gynecomastia?
- a) Spironolactone



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- b) Digoxin
- c) Cimetidine
- d) Amiloride

Q18. A newly introduced drug is under surveillance after being approved for public use. Data is being collected regarding rare and long-term adverse effects in a large population. Which phase of clinical trial does this represent?

- a) Phase I
- b) Phase II
- c) Phase III
- d) Phase IV

Q19. An 8-year-old child presents with episodes of staring spells along with occasional generalized tonic-clonic seizures. EEG confirms a mixed seizure disorder. The physician chooses a drug effective for multiple seizure types. Which of the following is most appropriate for long-term management?

- a) Ethosuximide
- b) Phenytoin
- c) Phenobarbital
- d) Valproate

Q20. A patient with osteoarthritis requires long-term NSAID therapy but has a history of peptic ulcer disease. The physician prefers a drug with minimal gastrointestinal toxicity while maintaining anti-inflammatory efficacy. Which of the following drugs is most appropriate?

- a) Aspirin
- b) Diclofenac
- c) Ibuprofen
- d) Celecoxib

Q21. A pharmacology experiment is conducted to study autonomic drug interactions. A subject is pretreated with an α -adrenergic blocker and subsequently administered adrenaline. Instead of the expected rise in blood pressure, a paradoxical fall in blood pressure is observed due to unopposed β 2-mediated vasodilation. This classical phenomenon is often demonstrated in experimental settings. Which of the following combinations best explains this response?

- a) β -blocker followed by dopamine
- b) α -blocker followed by adrenaline
- c) β -blocker followed by noradrenaline
- d) α -blocker followed by serotonin



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- Q22. A 25-year-old male presents to the emergency department with confusion, vomiting, abdominal pain, and rapid deep breathing. His breath has a characteristic fruity odor. Laboratory evaluation reveals hyperglycemia, metabolic acidosis, and ketonemia. A diagnosis of diabetic ketoacidosis is made. What is the most appropriate initial management for this patient?
- a) Sulfonylureas
 - b) Calcium gluconate
 - c) IV fluids with regular insulin
 - d) Ringer lactate alone
- Q23. A patient presents with chronic dyspepsia and is found to have *Helicobacter pylori* infection on diagnostic testing. The physician decides to initiate eradication therapy to reduce recurrence of peptic ulcer disease. A standard triple therapy regimen is planned. Which of the following combinations is most appropriate for this condition?
- a) Omeprazole + Amoxicillin + Clarithromycin
 - b) Metronidazole + Ciprofloxacin + Amoxicillin
 - c) Omeprazole + Erythromycin + Metronidazole
 - d) Metronidazole + Omeprazole + Ciprofloxacin
- Q24. A farmer is brought to the emergency department after a snake bite. He develops ptosis, difficulty in breathing, and progressive muscle weakness suggestive of neuromuscular paralysis. Antivenom is administered promptly. To further improve neuromuscular transmission, an additional pharmacologic agent is given. Which of the following is the most appropriate intervention?
- a) Pyridostigmine
 - b) Physostigmine
 - c) Neostigmine alone
 - d) Neostigmine with atropine
- Q25. A 32-year-old male is brought to the emergency department unconscious with shallow breathing and pinpoint pupils. His friend reports recent opioid consumption. On examination, he has respiratory depression and decreased level of consciousness. Immediate pharmacological reversal is required. Which of the following drugs should be administered?
- a) Flumazenil
 - b) Naltrexone
 - c) Naloxone
 - d) Atropine



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- Q26. A patient with type 2 diabetes mellitus and chronic kidney disease is planned for initiation of a DPP-4 inhibitor. The physician prefers a drug that does not require dose adjustment in renal impairment. Considering pharmacokinetic properties, which of the following drugs is most suitable?
- a) Saxagliptin
 - b) Sitagliptin
 - c) Alogliptin
 - d) Linagliptin
- Q27. A patient diagnosed with osteoporosis is started on pharmacotherapy. One of the prescribed drugs acts by stimulating osteoblastic activity, thereby increasing bone formation without significantly affecting bone resorption. Which of the following drugs has this unique mechanism of action?
- a) Alendronate
 - b) Raloxifene
 - c) Strontium ranelate
 - d) Teriparatide
- Q28. A 72-year-old male presents with urinary hesitancy, weak urinary stream, and increased PSA levels. He is diagnosed with benign prostatic hyperplasia. The prescribed drug works by reducing prostate size through inhibition of conversion of testosterone to dihydrotestosterone. Which of the following drugs is used in this condition?
- a) Tamsulosin
 - b) Prazosin
 - c) Finasteride
 - d) Bethanechol
- Q29. A patient with a history of severe motion sickness is planning a long sea voyage. He is prescribed a drug to prevent nausea and vomiting. The physician counsels him regarding possible anticholinergic adverse effects. Which of the following side effects is most likely to be experienced?
- a) Increased salivation
 - b) Diarrhea
 - c) Frequent urination
 - d) Dry mouth
- Q30. A woman presents with amenorrhea, galactorrhea, and visual disturbances. MRI reveals a pituitary adenoma compressing the optic chiasm. Serum prolactin levels are markedly elevated. She is diagnosed with prolactinoma. Which of the following drugs is considered the treatment of choice?
- a) Fluvastatin



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- b) Pramipexole
- c) Bromocriptine
- d) Octreotide

Q31. A diabetic patient on prolonged steroid therapy presents with facial pain and a black necrotic lesion over the nasal region. Imaging reveals invasive fungal infection involving the paranasal sinuses. A diagnosis of mucormycosis is made. Which of the following is the most appropriate initial antifungal therapy?

- a) Fluconazole
- b) Itraconazole
- c) Amphotericin B
- d) Griseofulvin

Q32. A young male presents with recurrent episodes of severe pain, anemia, and jaundice since childhood. Peripheral smear reveals sickled red blood cells. He is being evaluated for disease-modifying therapy. Which of the following drugs is NOT used in the management of this condition?

- a) Hydroxyurea
- b) Voxelotor
- c) Crizanlizumab
- d) Denosumab

Q33. In an enzyme kinetics study, a drug is found to compete with the substrate for binding at the active site of an enzyme. Increasing the substrate concentration reverses the inhibitory effect. This type of inhibition alters enzyme affinity but does not affect maximum velocity. Which of the following changes is expected?

- a) Decreased V_{max}
- b) Increased K_m
- c) Increased V_{max}
- d) Decreased K_m

Q34. A 76-year-old woman presents with progressive memory impairment, confusion, and inability to perform daily activities. MRI shows cortical atrophy. She is started on a monoclonal antibody targeting amyloid-beta plaques in the brain. Which of the following drugs is being administered?

- a) Denosumab
- b) Reslizumab
- c) Pertuzumab
- d) Aducanumab



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- Q35. A patient presents with acute hepatitis B infection. Laboratory findings show positive HBsAg and HBeAg along with elevated liver enzymes. Antiviral therapy is planned to reduce viral replication. Which of the following drugs is most appropriate in this condition?
- a) Zidovudine
 - b) Dolutegravir
 - c) Sofosbuvir
 - d) Lamivudine
- Q36. A patient with glaucoma develops cystoid macular edema after initiation of topical therapy. The adverse effect is attributed to increased uveoscleral outflow. Which of the following drugs is most likely responsible for this complication?
- a) Timolol
 - b) Brimonidine
 - c) Acetazolamide
 - d) Latanoprost
- Q37. A patient undergoing antitubercular therapy presents with tingling and numbness in the hands and feet. This adverse effect is attributed to pyridoxine deficiency caused by one of the drugs. Which of the following supplements should be administered to prevent and treat this condition?
- a) Vitamin B12
 - b) Thiamine
 - c) Pyridoxine
 - d) Folic acid
- Q38. A patient undergoing chemotherapy develops progressive shortness of breath. Imaging reveals features consistent with pulmonary fibrosis. The adverse effect is suspected to be drug-induced. Which of the following chemotherapeutic agents is most likely responsible?
- a) Cisplatin
 - b) Doxorubicin
 - c) Bleomycin
 - d) 5-Fluorouracil
- Q39. A 6-month-old infant presents with recurrent flexor spasms and developmental delay. EEG shows a characteristic hypsarrhythmia pattern. A diagnosis of infantile spasms is made. Which of the following is the drug of choice for treatment?
- a) Ethosuximide
 - b) Valproate



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- c) Vigabatrin
- d) ACTH

Q40. A patient on long-term NSAID therapy develops epigastric pain and is diagnosed with gastric ulcer. A drug is prescribed that mimics prostaglandins and enhances mucosal protection. Which of the following drugs is most appropriate in this case?

- a) Antacids
- b) H₂ receptor blockers
- c) Misoprostol
- d) Proton pump inhibitors

Q41. A patient with long-standing type 2 diabetes mellitus is on basal insulin therapy but continues to have persistently elevated postprandial blood glucose levels. The physician decides to add a rapid-acting insulin preparation before meals to achieve better glycemic control. Which of the following insulin analogs is most appropriate for controlling postprandial hyperglycemia?

- a) Insulin glargine
- b) Insulin detemir
- c) Insulin degludec
- d) Insulin aspart

Q42. A patient undergoing chemotherapy for malignancy develops thrombocytopenia, leading to increased risk of bleeding. The treating physician initiates a drug that stimulates megakaryocyte proliferation and platelet production. Which of the following drugs is most appropriate for this indication?

- a) Filgrastim
- b) Erythropoietin
- c) Iron dextran
- d) Oprelvekin

Q43. A hypertensive patient is diagnosed with bilateral renal artery stenosis during evaluation for resistant hypertension. The physician carefully reviews antihypertensive medications due to the risk of worsening renal function. Which of the following drugs should be avoided in this patient?

- a) Hydralazine
- b) Methyldopa
- c) Dopamine
- d) Enalapril



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- Q44. A patient with Parkinson's disease is started on dopaminergic therapy. After a few weeks, he develops symptoms of peripheral vasospasm, including cold extremities and reduced peripheral circulation. Which of the following drugs is most likely responsible for this adverse effect?
- a) Levodopa
 - b) Ropinirole
 - c) Entacapone
 - d) Bromocriptine
- Q45. A patient presents with dry skin, tachycardia, blurred vision, and urinary retention after accidental ingestion of seeds from a wild plant. On examination, pupils are dilated and bowel sounds are reduced. These findings are suggestive of anticholinergic toxicity. Which of the following is the most likely cause?
- a) Organophosphate poisoning
 - b) Opioid overdose
 - c) Datura ingestion
 - d) Physostigmine toxicity
- Q46. A drug used in metabolic disorders acts by activating PPAR- γ receptors, leading to increased insulin sensitivity in peripheral tissues. It is commonly prescribed in patients with insulin resistance. Which of the following conditions is primarily treated with this class of drugs?
- a) Graves disease
 - b) Hypertension
 - c) Hyperthyroidism
 - d) Diabetes mellitus
- Q47. A patient with chronic NSAID use develops a gastric ulcer. The physician plans long-term therapy to promote healing and reduce recurrence by effectively suppressing gastric acid secretion. Which of the following drugs is most effective for this purpose?
- a) Antacids
 - b) H₂ receptor blockers
 - c) Misoprostol
 - d) Proton pump inhibitors
- Q48. A patient receiving chemotherapy develops severe thrombocytopenia due to bone marrow suppression. A drug is administered that stimulates megakaryocyte growth and differentiation to increase platelet production. Which of the following drugs is most appropriate?
- a) Filgrastim
 - b) Erythropoietin



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- c) Iron dextran
- d) Oprelvekin

Q49. A patient with hypertension is found to have renal artery stenosis on imaging studies. The physician considers pharmacologic therapy but is cautious about drugs that may worsen renal perfusion. Which of the following drugs is contraindicated in this patient?

- a) Hydralazine
- b) Methyldopa
- c) Dopamine
- d) Enalapril

Q50. A patient with Parkinson's disease develops symptoms of peripheral vasospasm after initiation of therapy. The drug responsible is known to act as a dopamine agonist with ergot derivative properties. Which of the following drugs is most likely responsible?

- a) Levodopa
- b) Entacapone
- c) Bromocriptine
- d) Ropinirole

Q51. A 23-year-old male presents with agitation, dry mouth, tachycardia, urinary retention, and blurred vision after ingesting seeds from a wild plant. On examination, his skin is flushed and dry. These findings are consistent with an anticholinergic toxidrome. Which of the following is the most likely cause?

- a) Organophosphate poisoning
- b) Opioid overdose
- c) Physostigmine toxicity
- d) Datura ingestion

Q52. A 60-year-old male with chronic heart failure presents with progressive dyspnea and fatigue. Echocardiography reveals reduced ejection fraction. The physician prescribes a drug known to reduce mortality by inhibiting sympathetic overactivity and preventing cardiac remodeling. Which of the following drugs is most appropriate?

- a) Dobutamine
- b) Furosemide
- c) Milrinone
- d) Carvedilol



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- Q53. A hypertensive patient is started on an ACE inhibitor. After a few weeks, he develops persistent dry cough and mild hyperkalemia. The physician explains these are common adverse effects of the drug. Which of the following is NOT associated with ACE inhibitor therapy?
- a) Cough
 - b) Hyperkalemia
 - c) Angioedema
 - d) Hypokalemia
- Q54. A newly developed antiarrhythmic drug is found to prolong the duration of the cardiac action potential and refractory period without affecting the rate of depolarization. ECG shows prolonged QT interval. This pharmacologic profile corresponds to which class of antiarrhythmic drugs?
- a) Class I
 - b) Class II
 - c) Class III
 - d) Class IV
- Q55. A patient undergoing chemotherapy develops features of tumor lysis syndrome, including hyperuricemia, hyperkalemia, and acute kidney injury. A drug is administered that rapidly converts uric acid into a more soluble metabolite for excretion. Which of the following drugs is most appropriate?
- a) Febuxostat
 - b) Colchicine
 - c) Rasburicase
 - d) Alendronate
- Q56. A patient presents with fever, cough, diarrhea, and confusion after exposure to contaminated water from an air-conditioning system. Chest imaging shows patchy infiltrates. Legionella infection is suspected. Which of the following is the drug of choice?
- a) Norfloxacin
 - b) Daptomycin
 - c) Ceftibiprole
 - d) Azithromycin
- Q57. A patient with bronchial asthma presents with an acute exacerbation and is started on bronchodilators. One of the prescribed drugs has a slower onset of action and is not suitable for immediate relief of acute symptoms. Which of the following drugs is least appropriate for acute attacks?
- a) Salbutamol
 - b) Terbutaline



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- c) Formoterol
- d) Salmeterol

Q58. A 25-year-old male presents with sudden onset palpitations and dizziness. ECG reveals supraventricular tachycardia. A rapid-acting intravenous drug is administered, causing transient AV nodal block and immediate termination of arrhythmia. Which of the following drugs was used?

- a) Verapamil
- b) Amiodarone
- c) Lidocaine
- d) Adenosine

Q59. A patient develops excessive bleeding following thrombolytic therapy with streptokinase for myocardial infarction. An antifibrinolytic drug is administered to counteract this effect. Which of the following drugs is most appropriate?

- a) Vitamin K
- b) Noradrenaline
- c) Tranexamic acid
- d) Rutin

Q60. A patient presents with nausea and vomiting and is prescribed an antiemetic that acts as a dopamine receptor antagonist and also enhances gastrointestinal motility via 5-HT₄ receptor agonism. Which of the following drugs is most likely used?

- a) Ondansetron
- b) Ibutilide
- c) Benzhexol
- d) Metoclopramide

Q61. A patient with bronchial asthma on long-term inhaled corticosteroid therapy develops oral candidiasis, presenting with white patches in the oral cavity. To minimize systemic side effects and reduce local adverse effects, the physician switches to a prodrug that is activated within the lungs. This drug has minimal systemic bioavailability due to rapid inactivation in circulation. Which of the following is the most appropriate choice?

- a) Budesonide
- b) Fluticasone
- c) Mometasone
- d) Ciclesonide



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- Q62. A cancer patient undergoing chemotherapy develops severe nausea and vomiting refractory to standard treatment. A drug is administered that acts by blocking serotonin receptors in the chemoreceptor trigger zone and gastrointestinal tract. This agent is particularly effective in chemotherapy-induced emesis. Which of the following drugs is most appropriate?
- a) Domperidone
 - b) Scopolamine
 - c) Doxylamine
 - d) Palonosetron
- Q63. A patient receiving treatment for nausea develops acute dystonia, restlessness, and muscle rigidity. These extrapyramidal symptoms are attributed to dopamine receptor blockade in the central nervous system. Which of the following drugs is most likely responsible for these adverse effects?
- a) Ondansetron
 - b) Doxylamine
 - c) Domperidone
 - d) Metoclopramide
- Q64. A chemotherapeutic agent used in the treatment of leukemia acts specifically during the S-phase of the cell cycle by inhibiting DNA synthesis. It interferes with DNA polymerase activity, thereby preventing replication. Which of the following drugs best fits this mechanism of action?
- a) Cyclophosphamide
 - b) Vincristine
 - c) Cytarabine
 - d) Vinblastine
- Q65. A patient undergoing chemotherapy develops painful erythema, swelling, and desquamation of the palms and soles. This presentation is consistent with hand-foot syndrome, a known adverse effect of certain chemotherapeutic agents. Which of the following drugs is most commonly associated with this condition?
- a) Cyclophosphamide
 - b) Chloramphenicol
 - c) Mitomycin C
 - d) Capecitabine
- Q66. A patient with muscle spasticity is treated with botulinum toxin injections. The toxin produces muscle relaxation by interfering with neurotransmitter release at the neuromuscular junction. Which of the following best describes its mechanism of action?
- a) Stimulates acetylcholine receptors



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- b) Inhibits acetylcholinesterase
- c) Blocks calcium entry into nerve terminals
- d) Inhibits acetylcholine release

Q67. A patient presents with an urge to move his legs at night, accompanied by discomfort that is relieved by movement. He is diagnosed with restless legs syndrome. Laboratory evaluation reveals iron deficiency anemia. Which of the following is the most appropriate first-line therapy in this case?

- a) Gabapentin
- b) Tramadol
- c) Iron supplementation
- d) Vitamin B12

Q68. A patient treated for bacterial meningitis develops right upper quadrant pain. Imaging studies reveal biliary sludge and gallstones. This is a recognized adverse effect of a commonly used antibiotic. Which of the following drugs is most likely responsible?

- a) Penicillin
- b) Ampicillin
- c) Vancomycin
- d) Ceftriaxone

Q69. A patient with heart failure following myocardial infarction presents with dyspnea and orthopnea. The physician initiates a drug known to reduce mortality by preventing cardiac remodeling and decreasing afterload. Which of the following drugs is most appropriate for long-term management?

- a) Hydrochlorothiazide
- b) Terazosin
- c) Amlodipine
- d) Lisinopril

Q70. A patient presents with episodic headache, palpitations, and sweating. Blood pressure is found to be 180/110 mmHg. Investigations confirm pheochromocytoma. Prior to surgical removal of the tumor, pharmacological preparation is required to prevent hypertensive crises. Which of the following drugs should be administered first?

- a) Labetalol
- b) Nitroprusside
- c) Phentolamine
- d) Phenoxybenzamine



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- Q71. A patient with mechanical heart valves develops atrial fibrillation. Long-term anticoagulation is required to prevent thromboembolic complications. Considering efficacy and safety in this condition, which of the following drugs is most appropriate?
- a) Direct oral anticoagulants
 - b) Alteplase
 - c) Heparin
 - d) Warfarin
- Q72. A patient develops severe diarrhea after receiving broad-spectrum antibiotics. *Clostridioides difficile* infection is suspected. Appropriate antimicrobial therapy is initiated. Which of the following drugs is NOT used in the treatment of this condition?
- a) Vancomycin
 - b) Metronidazole
 - c) Fidaxomicin
 - d) Ciprofloxacin
- Q73. A patient with organophosphate poisoning is treated with atropine and pralidoxime. The latter drug acts by restoring enzyme activity that has been inhibited by the toxin. Which of the following best describes the mechanism of action of pralidoxime?
- a) Inhibits acetylcholine release
 - b) Blocks muscarinic receptors
 - c) Reactivates acetylcholinesterase
 - d) Inhibits choline uptake
- Q74. A patient undergoing chemotherapy develops severe vomiting. A drug is administered that acts centrally by blocking serotonin receptors in the chemoreceptor trigger zone. This mechanism is particularly effective in chemotherapy-induced nausea. Which of the following mechanisms best describes this action?
- a) D2 receptor blockade
 - b) H1 receptor blockade
 - c) Muscarinic receptor blockade
 - d) 5-HT₃ receptor blockade
- Q75. A postoperative patient receiving opioid analgesia develops severe biliary colic. This is attributed to increased tone of the sphincter of Oddi caused by the drug. Which of the following agents is most likely responsible for this adverse effect?



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- a) Naloxone
- b) Atropine
- c) Morphine
- d) Aceclofenac

Q76. A patient with hypertension and benign prostatic hyperplasia presents with urinary hesitancy and weak stream. A drug is prescribed that improves urinary flow by selectively blocking α_1A receptors in the prostate, with minimal systemic hypotensive effects. Which of the following is most appropriate?

- a) Terazosin
- b) Finasteride
- c) Tamsulosin
- d) Phenoxybenzamine

Q77. A patient with chronic obstructive pulmonary disease is prescribed tiotropium, a long-acting anticholinergic bronchodilator. However, caution is advised in certain comorbid conditions due to its anticholinergic effects. In which of the following conditions is this drug contraindicated?

- a) Asthma
- b) Hypertension
- c) Peptic ulcer disease
- d) Urinary retention

Q78. A patient with acute decompensated heart failure is treated with nesiritide, a recombinant B-type natriuretic peptide analog. It produces vasodilation and natriuresis. Which of the following statements regarding this drug is incorrect?

- a) Causes natriuresis
- b) Used in acute CHF
- c) Given orally
- d) BNP analogue

Q79. A patient is prescribed imipenem for a severe infection. It is co-administered with cilastatin to prevent its degradation in the kidneys. Which of the following mechanisms explains this pharmacological interaction?

- a) Beta-lactamase inhibition
- b) Increased drug absorption
- c) Synergistic antibacterial action
- d) Inhibition of renal dehydropeptidase



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- Q80. A patient on allopurinol therapy for gout requires dose adjustment of another drug due to a significant drug interaction. Allopurinol inhibits xanthine oxidase, affecting the metabolism of certain medications. Which of the following drugs requires dose reduction when used concomitantly?
- Cyclophosphamide
 - 5-Fluorouracil
 - Cimetidine
 - 6-Mercaptopurine
- Q81. A female patient with epilepsy on long-term antiepileptic therapy presents with gum hypertrophy, excessive hair growth, and coarsening of facial features. She reports no recent change in medication. On examination, gingival overgrowth is evident. Which of the following drugs is most likely responsible for these adverse effects?
- Carbamazepine
 - Valproate
 - Ethosuximide
 - Phenytoin
- Q82. A patient undergoes extensive jejunal resection and subsequently develops malabsorption with chronic diarrhea and weight loss. A drug is prescribed that enhances intestinal mucosal growth and improves nutrient absorption. Which of the following drugs is most appropriate for this condition?
- Semaglutide
 - Exenatide
 - Liraglutide
 - Teduglutide
- Q83. A pregnant woman in her first trimester presents with fever and chills. Peripheral smear confirms infection with *Plasmodium vivax*. Considering safety for both mother and fetus, an antimalarial drug is chosen. Which of the following is the most appropriate treatment?
- Artemisinin-based combination therapy
 - Doxycycline
 - Chloroquine
 - Primaquine
- Q84. A patient presents with CNS depression, respiratory compromise, and metabolic acidosis following drug overdose. He is managed with sodium bicarbonate therapy. The toxicological profile suggests overdose of a drug causing respiratory depression and acidosis. Which of the following is most likely responsible?
- Morphine



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- b) Amphetamine
- c) Phenobarbitone
- d) Opioids

Q85. A patient with benign prostatic hyperplasia is started on a drug that reduces prostate size by blocking conversion of testosterone to dihydrotestosterone. This leads to gradual improvement in urinary symptoms. Which of the following enzymes is inhibited by this drug?

- a) Alpha receptor
- b) Beta receptor
- c) Androgen receptor
- d) 5-alpha reductase

Q86. A patient on long-term lithium therapy for bipolar disorder develops excessive urination and thirst. Laboratory evaluation suggests nephrogenic diabetes insipidus. The physician prescribes a drug that reduces lithium entry into renal tubular cells. Which of the following is most appropriate?

- a) Vasopressin
- b) Diclofenac
- c) Indapamide
- d) Amiloride

Q87. A patient presents with digoxin toxicity characterized by arrhythmias, nausea, and visual disturbances such as yellow vision. Immediate intervention is required to prevent fatal complications. Which of the following is the most effective treatment?

- a) Atropine
- b) Cardioversion
- c) Hemodialysis
- d) Digoxin-specific antibody fragments

Q88. A drug's maintenance dose is determined based on pharmacokinetic parameters to maintain steady-state concentration. It primarily depends on the rate at which the drug is eliminated from the body. Which of the following parameters is most important in determining maintenance dose?

- a) Volume of distribution
- b) Bioavailability
- c) Clearance
- d) Loading dose



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- Q89. A patient with arrhythmia is treated with a beta-blocker that reduces heart rate and slows AV nodal conduction. According to the Vaughan Williams classification, this drug belongs to which class of antiarrhythmics?
- a) Quinidine – Class IB
 - b) Amiodarone – Class IV
 - c) Lidocaine – Class IA
 - d) Esmolol – Class II
- Q90. An outbreak of meningococcal meningitis occurs in a community, and mass chemoprophylaxis is initiated for close contacts. The drug chosen should effectively eradicate nasopharyngeal carriage of the organism. Which of the following drugs is most appropriate?
- a) Ciprofloxacin
 - b) Doxycycline
 - c) Rifampicin
 - d) Chloramphenicol
- Q91. A drug is studied for its mechanism of action and found not to interfere with nucleic acid synthesis. Instead, it inhibits protein synthesis by acting on the ribosomal subunit. Which of the following drugs fits this profile?
- a) Rifampicin
 - b) Fluoroquinolones
 - c) Nitrofurantoin
 - d) Linezolid
- Q92. A patient with myasthenia gravis is prescribed antibiotics for an infection. Certain drugs are known to worsen neuromuscular transmission and exacerbate symptoms. Which of the following should be avoided in this patient?
- a) Penicillin
 - b) Sulfonamides
 - c) Aminoglycosides
 - d) All of the above
- Q93. A patient on long-term antipsychotic therapy presents with fever, sore throat, and severe neutropenia. Laboratory findings suggest agranulocytosis. Which of the following drugs is most likely responsible for this adverse effect?
- a) Risperidone
 - b) Olanzapine



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- c) Clozapine
- d) Pimozide

Q94. A patient with congestive heart failure is on diuretics and develops hypokalemia. A potassium-sparing diuretic is added to correct the electrolyte imbalance. This drug acts on which segment of the nephron?

- a) Proximal convoluted tubule
- b) Loop of Henle
- c) Distal convoluted tubule
- d) Collecting duct

Q95. A drug is administered that does not cross the blood-brain barrier due to its quaternary ammonium structure. It is used for its peripheral anticholinergic effects without central side effects. Which of the following drugs fits this description?

- a) Atropine
- b) Hyoscine
- c) Physostigmine
- d) Glycopyrrolate

Q96. A patient on monoamine oxidase inhibitors consumes cheese and develops severe hypertension due to tyramine accumulation. Immediate treatment is required to control blood pressure. Which of the following drugs is most appropriate?

- a) Prazosin
- b) Pentazocine
- c) Phentolamine
- d) Phenoxybenzamine

Q97. A newborn presents with limb deformities characterized by phocomelia. On history, the mother had taken a drug during early pregnancy. Which of the following drugs is most likely responsible for this teratogenic effect?

- a) Tetracycline
- b) Warfarin
- c) Alcohol
- d) Thalidomide

Q98. A patient with heart failure is started on a drug combination that increases natriuretic peptides and reduces cardiac remodeling by inhibiting neprilysin and blocking angiotensin receptors. Which of the following drugs is used in this therapy?

- a) Enalapril
- b) Losartan
- c) Spironolactone
- d) Sacubitril + Valsartan



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Q99. A patient with type 2 diabetes mellitus is started on a drug that promotes glucose excretion through urine by inhibiting renal glucose reabsorption. After a few weeks, he develops recurrent genital fungal infections. Which class of drugs is responsible?

- a) Sulfonylureas
- b) DPP-4 inhibitors
- c) Thiazolidinediones
- d) SGLT-2 inhibitors

Q100. A patient with tuberculosis is started on multidrug therapy. After a few days, he notices orange discoloration of urine, sweat, and tears. This is explained as a known harmless side effect of one of the drugs. Which of the following is responsible?

- a) Isoniazid
- b) Ethambutol
- c) Pyrazinamide
- d) Rifampicin

ANSWERS

1	d
2	c
3	c
4	b
5	c
6	d
7	d
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9	c
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99	d
100	d